Portfolio

Questions and Answers

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Table of Contents

Long Question (Attempted: Q1)

Q1. Explain how unsupervised learning can be applied in a large-scale digital payment platform like eSewa.

Unsupervised learning forms an approach in machine learning which is used for processing unlabeled data to extract any kind of essential structure that exists in the data. It does not require any predetermined output, unlike supervised learning. Thus, it is more efficient when it’s a case of applications which involve millions of transactions everyday as well as labeling of the data becomes difficult. Additionally, unsupervised learning helps in knowing consumer behaviors.

Customer Segmentation is considered as one of the main examples of unsupervised learning. Consumers of a particular digital payment gateway differ in the manner of using the services provided. Some people may perform transactions regularly such as paying for their bills as well as topping up their accounts while others may perform transactions occasionally. Using k-means algorithm, people may be segmented on the basis of frequency of transactions, money spent and the type of services that are used. Consumers of a digital payment gateway who perform transactions frequently and those who transact rarely but pay their bills only are some of the clusters formed by customer segmentation. This allows organizations to target consumers specifically through marketing. The companies may offer discounts, cashback or programs for their clients.

A further application involves detecting frauds. Typically, dishonest actions in digital payments have unique patterns when compared to normal user actions. Because the number of labeled fraudulent examples is limited, unsupervised techniques like Isolation Forest prove useful in spotting such oddities. In other words, an action that is not common, such as making a large amount of payments from an unusual place, could be recognized as potential fraud by the system. The user could get a notification about this kind of situation, or the transaction would not be done for verification. Such approaches might also result in false positives.

To conclude, unsupervised learning is essential for helping digital payment services like eSewa learn from user actions and improve safety without labels.

Short Questions

Topic: Overfitting / Model Issues

Q1. Overfitting vs Underfitting

Both overfitting and underfitting are typical issues in machine learning. Overfitting occurs when a model trains too much and includes all details such as noise. Thus, overfitted models can work perfectly on training data but fail to predict in an accurate way for any new test data. For example, a model can memorize all training examples rather than finding general rules.

On the other hand, underfitting refers to a situation where a model cannot learn even the basic rules of the training dataset due to its complexity. Underfitting occurs because a simple model is used by machine learning algorithm to represent data. In other words, underfitting is just the opposite of overfitting.

The key difference is that overfitting occurs because a model becomes too much complicated with high variance. At the same time, underfitting is attributed to a simple model having a low variance but high bias. It can be mentioned that both overfitting and underfitting have negative effects on the quality of the model.

Topic: RNN / LSTM

Q1. RNN vs LSTM

Recurrent neural networks are specialized neural networks used for modeling sequential information such as texts or sounds. RNN models work by storing a history of previous inputs. However, recurrent neural networks cannot process longer inputs as they face the problem of vanishing gradients.

The Long Short-Term Memory Network (LSTM), is a variant of RNNs. It helps in solving the mentioned problem. This network introduces specific components called gates that are input gates, forget gates, and output gates, which allow the model to determine what information should be stored and what to ignore. For instance, in the case of a text, LSTM can store important information regarding some word at the beginning while reading subsequent words of the sentence.

So, we can conclude that, LSTM models perform better than RNN in case if the model faces a task which requires it to maintain long-term dependencies, such as translations and sound recognition. Even though LSTM is more complicated, it works better than RNN.